

Neuroticism and Amygdala/dmPFC Activation During Dynamic Face Perception

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Abstract

The present study investigated whether trait neuroticism is associated with altered activation of the amygdala and dorsomedial prefrontal cortex (dmPFC) during dynamic emotional face perception. A sample of 196 healthy young adults from the PIOP1 dataset of the Amsterdam Open MRI Collection viewed short video clips depicting anger, contempt, joy, pride, and neutral expressions. First-level general linear models with separate regressors for each expression type were estimated using preprocessed BOLD data and an extensive set of motion and physiological nuisance regressors. Four contrasts (all emotions versus neutral, negative versus neutral, positive versus neutral, and negative versus positive) were entered into second-level regression models testing linear associations with NEO-FFI neuroticism scores, while controlling for age, sex, and head motion. Neither region-of-interest analyses in bilateral amygdala and dmPFC, nor whole-brain voxelwise FDR-corrected analyses revealed significant associations between neuroticism and BOLD responses for any contrast. Exploratory uncorrected maps showed only scattered, small clusters. These findings indicate that, in this sample and task context, neuroticism does not exhibit robust linear relationships with regional amygdala or dmPFC activation to emotional faces. This suggests that neuroticism-related differences in emotional face processing may be subtle, context-dependent, or more evident at the level of network connectivity than localized mean activation.

Keywords: neuroticism, emotional face processing, amygdala, dorsomedial prefrontal cortex, AOMIC

Github: <https://github.com/WarreLinseele/Case-Studies-MRI>

Dataset: <https://openneuro.org/datasets/ds002785/versions/2.0.0>

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Neuroticism and Emotional Face Processing

Neuroticism, a core dimension of the Five-Factor Model of personality, is characterized by a stable tendency toward negative affect, emotional instability, and heightened sensitivity to potentially aversive stimuli (Eysenck & Eysenck, 1985; Meiering et al., 2026). Because neuroticism is a robust transdiagnostic risk factor for mood and anxiety disorders, understanding how it shapes basic affective processes such as the perception of emotional faces may help explain the cognitive and neural pathways linking personality to psychopathology (Ametaj et al., 2015; Griffith et al., 2009; Struijs et al., 2021). Behavioral studies suggest that neuroticism is associated with biased processing of emotional facial expressions. These biases differ for positive versus negative expressions.

Starting with positive expressions, Andric et al. (2016) found that healthy adults higher in neuroticism were less accurate at recognizing happy facial expressions, pointing to a selective insensitivity to positive emotional signals. Similarly, Sawada et al. (2016) reported that highly neurotic participants were slower to detect happy facial expressions and showed greater subjective arousal in response to them. Consistent with these findings, Rappaport et al. (2018) also found evidence of reduced recognition of positive emotional expressions among individuals high in neuroticism.

In contrast, research suggests that neuroticism is associated with heightened sensitivity to negative emotional information. Chan et al. (2007) showed that never-depressed individuals high in neuroticism processed negative information more quickly and positive information less quickly across several emotional tasks, including facial expression recognition. Likewise, Hopkins et al. (2021) found that individuals high in neuroticism required less emotional intensity to classify a face as fearful or angry, indicating heightened sensitivity specifically to negative facial cues. Taken together, these findings suggest that

neuroticism is associated with differential processing of positive and negative facial expressions. However, it remains unclear how these effects are reflected at the neural level and whether they follow a similar pattern to that observed at the behavioral level.

Amygdala and Facial Expressions

At the neural level, the amygdala plays a central role in evaluating the emotional and social significance of faces, integrating sensory information with affective relevance (Šimić et al., 2021). Several functional imaging studies link neuroticism, or the closely related construct of anxiety-proneness, to heightened amygdala reactivity to facial stimuli. Stein et al. (2007) found that anxiety-prone young adults, selected partly on the basis of elevated neuroticism, showed greater bilateral amygdala activation during an emotional-face assessment task than less anxiety-prone peers. Similarly, Chan et al. (2009) reported that healthy volunteers high in neuroticism exhibited stronger right amygdala responses to fearful relative to happy facial expressions.

However, evidence for an association between neuroticism and amygdala reactivity to negative faces remains inconsistent. Klamer et al. (2017), using a whole-brain analysis, did not observe amygdala involvement in relation to neuroticism during emotional face processing. Similarly, in a large twin sample, Silverman (2018) found no reliable association between neuroticism and amygdala activation or amygdala habituation during a negative emotional face-matching task. This discrepancy may reflect differences in task design, such as static versus dynamic visual stimuli, blocked versus event-related paradigms, or the specific emotions used.

Positive expressions have received less empirical attention. Stein et al. (2007) found that the heightened bilateral amygdala activation observed in anxiety-prone individuals was not limited to negative stimuli but extended to positive facial expressions as well. This finding is inconsistent with both the whole-brain results of Klamer et al. (2017) and the

behavioral literature reviewed above, which generally suggests reduced sensitivity and recognition accuracy for positive expressions among individuals high in neuroticism (Andric et al., 2016; Sawada et al., 2016; Rappaport et al., 2018). Accordingly, the direction of the association between neuroticism and amygdala responses to positive facial expressions remains unclear.

dmPFC and Facial Expressions

In addition to the amygdala, the dorsomedial prefrontal cortex (dmPFC) has been implicated in the processing of emotional facial expressions and is therefore a key region of interest in the present study. The dmPFC is involved in the cognitive evaluation, interpretation, and regulation of socioemotional information, particularly when facial expressions are negative, ambiguous, or self-relevant (Ebner et al., 2012). Several studies have linked neuroticism to altered dmPFC engagement during emotional processing. Cremers et al. (2010) reported that higher neuroticism was associated with greater dmPFC activation in response to fearful faces. The authors further found stronger functional connectivity between the right amygdala and dmPFC as neuroticism increased, suggesting increased processing of negative emotional information among highly neurotic individuals. Similarly, Brown et al. (2020) observed that higher neuroticism predicted greater right dmPFC activation when participants attended to negative facial expressions. Evidence from Deng et al. (2019) further indicated that neuroticism was associated with stronger amygdala–dmPFC connectivity during the processing of negative emotional stimuli.

However, these findings are not entirely consistent. Meiering et al. (2026) found no association between neuroticism and regional brain activity during a negative emotional face-classification task. Instead, neuroticism was related to altered functional connectivity across a broader network including the amygdala, dorsolateral prefrontal cortex, anterior insula, and dorsal cingulate cortex. In addition, Klamer et al. (2017) did not identify the dmPFC in their

whole-brain analysis as a region associated with neuroticism during emotional face viewing. Despite these inconsistencies, several studies report increased dmPFC responses to negative facial expressions and increased amygdala-dmPFC connectivity during negative face processing with higher neuroticism.

Evidence linking neuroticism to dmPFC responses to positive facial expressions is more limited than for negative expressions. Among the few studies examining positive stimuli, Britton et al. (2007) reported that neuroticism was positively associated with rostral dmPFC activation to positive emotional stimuli, including facial expressions, although this effect was not specific to faces. In contrast, Haas et al. (2008) found no association between neuroticism and sustained neural responses to happy facial expressions. Overall, findings do not consistently indicate a clear link between neuroticism and dmPFC responses to positive facial expressions.

Present Study

Current literature suggests that neuroticism is associated with altered amygdala and dmPFC engagement during facial emotion processing. However, findings regarding the direction, valence-specificity, and whether these effects are regional or connectivity-based remain inconsistent across studies. The present study addresses this gap using functional MRI data from the Amsterdam Open MRI Collection (AOMIC), specifically the PIOP1 dataset (Snoek et al., 2021), which includes face-perception fMRI data and personality measures from over 200 healthy adults. During the task, participants viewed dynamic facial expressions of positive, negative, and neutral emotions, allowing for a direct comparison of valence effects within the same sample. This design allows us to examine whether neuroticism is associated with altered amygdala and dmPFC activity in response to emotional faces in general. It also enables us to directly test whether any associations differ systematically as a function of emotional valence.

Accordingly, the present study tests four hypotheses that move from this general question toward increasingly valence-specific predictions. First, higher neuroticism scores are expected to be associated with increased amygdala and dmPFC activity for emotionally loaded facial expressions relative to neutral expressions, consistent with reports of generally heightened amygdala reactivity to emotional faces among highly neurotic individuals. Second, higher neuroticism scores are expected to be associated with increased amygdala and dmPFC activity specifically for negative facial expressions, in line with evidence that neuroticism is tied to heightened sensitivity and self-referential processing of negative facial information. Third, given the more mixed associations reported for positive stimuli in prior work, higher neuroticism scores are expected to be associated with altered activity in the amygdala and dmPFC for positive facial expressions, though no specific direction of this association is hypothesized. Finally, the association between neuroticism and amygdala/dmPFC activity is expected to be stronger for negative than for positive facial expressions, reflecting a valence-specific rather than a general-sensitivity account of neuroticism.

Methods

Data Sample

Data were obtained from the Amsterdam Open MRI Collection (AOMIC), a set of publicly available multimodal MRI datasets designed to enable robust individual-differences analyses. The collection consists of three 3T MRI datasets, ID1000, PIOP1, and PIOP2, with structural, diffusion-weighted, resting-state, and task-based function MRI, as well as demographic and psychometric measures. The current study utilizes the PIOP1 dataset, which includes task-based fMRI acquisitions for several paradigms, and extensive personality and behavioral data including NEO-FFI neuroticism scores. All data are organized in BIDS format. Preprocessed derivatives, including fMRIPrep outputs, are publicly available.

The PIOP1 dataset contains data from 216 healthy university students recruited from the University of Amsterdam and the Amsterdam University of Applied Sciences. Participants were between approximately 18.25 and 26.25 years old (mean = 22.18, SD = 1.80). Biological sex was predominantly female (56% female). Self-reported handedness was primarily right-handed (n = 180 right-handed, 24 left-handed, 5 ambidextrous, 7 missing). Eligibility was assessed using a standard MRI safety checklist. All participants provided written informed consent in accordance with the local ethics committee (PIOP1 EC number 2015-EXT-4366). Of the 216 participants in the PIOP1 dataset, 203 met the motion quality-control criteria (mean framewise displacement ≤ 0.5 mm and spike fraction $\leq 20\%$) and had usable first-level fMRI data. A further seven participants were excluded due to incomplete phenotype data (missing neuroticism scores, age, or sex), yielding a final sample of $N = 196$. These 196 participants provided data for all four contrasts in the region-of-interest (ROI) and whole-brain regression analyses.

All MRI data were acquired on a Philips 3T Achieva scanner (Philips Healthcare, The Netherlands) using a 32-channel head coil, as part of the PIOP1 study. Functional data for the face perception task were collected using a T2*-weighted gradient-echo EPI sequence with multiband acceleration factor 3. For the run during the face perception task, 330 volumes were acquired with a repetition time (TR) of 0.75 seconds, yielding a total scan duration of 247.5 seconds. Other sequence parameters followed the standard PIOP1 protocol (Snoek et al., 2021), including whole-brain coverage and isotropic voxels. The functional slice stack was not angled relative to the anatomical scans.

fMRI Face Perception Task

We analyzed the PIOP1 face perception task, which was designed to measure neural responses to dynamic emotional face expressions. The task used an event-related design with separately randomized trial order for each participant. On each trial, participants passively

viewed a short dynamic video of 2 seconds from the Amsterdam Dynamical Facial Expression Set (ADFES), depicting one of five trial types: anger, contempt, joy, pride, or neutral. Emotional trials showed a transition from a neutral face to a full emotional expression, while neutral trials showed a static face with no facial movements. Each clip was followed by a fixed inter-stimulus interval of 5 seconds during which a blank screen with fixation point was shown.

Each expression was presented 6 times, with different actors for each presentation. In addition, one randomly selected emotion was presented three extra times. Extra trials always used the same actor and always appeared as the first three trials to enable evaluation of repetition-related effects. All faces were females with either North-European or Mediterranean background. The total task duration was 247.5 seconds (330 volumes at TR = 0.75 seconds). Trial onset, duration, expression, and model identity and ethnicity are encoded in the BIDS event files for the face perception task.

Preprocessing of MRI Data

Raw data were available in BIDS and had been preprocessed using the AOMIC pipeline based on fMRIPrep (version 1.4.1). For each participant, anatomical T1-weighted scans were corrected for intensity non-uniformity, skull-stripped, and segmented into tissue classes. Afterwards, images were non-linearly registered to the ICBM 152 Nonlinear Asymmetrical 2009c (MNI152NLin2009cAsym) template using ANTs. Cortical surfaces and anatomical parcellations were obtained using the workflow for FreeSurfer. Functional BOLD images for the face perception task were corrected for motion using rigid-body realignment, and for susceptibility distortion using a fieldmap-less approach. Functional images were co-registered to the T1-weighted image and normalized to MNI152NLin2009cAsym space using a single interpolation step. Slice-timing correction was not applied to the functional data. fMRIPrep produced subject-specific brain masks and confound time series for each run.

For the current analyses, we built a nuisance-regressor set from the fMRIPrep confound files for each face-perception run. From each confound file, we selected a 24-parameter motion model comprising the six rigid-body realignment parameters (three translations and three rotations), their temporal derivatives, and the squared terms of both the original and derivative parameters (Friston 24 model). We included the first five anatomical CompCor components from fMRIPrep, capturing variance in noise-dominated white-matter and cerebrospinal fluid regions. Non-steady-state volumes identified by fMRIPrep were added as separate indicator regressors. We included one binary spike regressor per high-motion volume, so that each flagged volume was modelled by a dedicated nuisance regressor in the design matrix. All the selected confound columns were concatenated into a single confound matrix per run. Any remaining missing values in these regressors, which were rare, were set to zero. Columns with more than one missing value were logged but not removed, as these typically reflected isolated omitted entries in valid regressors.

Additionally, we applied motion-based quality control to exclude excessively noisy subjects. For each participant, mean framewise displacement (FD, millimeters) was computed from the fMRIPrep confound time series, and the proportion of volumes with large motion was estimated using the spike regressors. Participants were excluded from analyses if their mean FD exceeded 0.5 millimeters or if more than 20% of volumes in the face perception task run were flagged as motion outliers. Subject-specific mean FD was kept as a continuous covariate and added to the second-level regression models to limit the risk of systematic associations between neuroticism and head motion to inflate brain-behaviour correlations.

First-level GLM

We estimated task-related BOLD responses for each participant using a mass-univariate first-level general linear model (GLM) implemented in the FirstLevelModel package of Nilearn. The models were fitted to the preprocessed face-perception BOLD time

series in MNI152NLin2009cAsym space, with a repetition time of 0.75 seconds to match the acquisition. For each subject, we built a design matrix with five task regressors: anger, contempt, joy, pride, and neutral. Onsets and durations for these regressors were taken from the BIDS event files. All events had a fixed duration of 2 seconds, while the 5-second inter-stimulus intervals were treated as implicit baseline. Task regressors were convolved with the canonical SPM double-gamma haemodynamic response function, and slow drifts were modeled with a cosine basis set using a high-pass cutoff of 1/128 Hz.

To account for structure of noise in the time series, we modeled temporal autocorrelation with a first-order autoregressive process. We applied spatial smoothing with a 6 millimeter full-width at half-maximum Gaussian kernel, and used the subject-specific brain mask produced by fMRIPrep to restrict the analysis to brain voxels. The nuisance regressors derived from the confound files, namely the 24-parameter motion model, aCompCor components, non-steady-states indicators, and spike regressors, were entered as additional regressors in the first-level design matrix for each participant.

From each fitted model, we computed four contrasts reflecting different emotion-related effects. **Our first hypothesis was defined as the average of the four emotional expressions (anger, contempt, joy and pride) minus the neutral expression.** Our second hypothesis was tested using a contrast that averaged anger and contempt, and subtracted neutral. Similarly, the contrast for the third hypothesis averaged joy and pride and subtracted neutral. The contrast for the fourth hypothesis compared the negative and positive averages. For each contrast and each participant, we computed effect-size beta images and corresponding z-statistic maps in NIfTI format.

Regions of Interest

Region-of-interest (ROI) analyses focused on bilateral amygdala and dorsomedial prefrontal cortex (dmPFC), consistent with earlier studies linking neuroticism to threat-

related limbic and medial prefrontal activity. Amygdala ROIs were anatomically defined using the Harvard-Oxford subcortical atlas, with 2 millimeter resolution and 25% threshold. We extracted binary masks for the left amygdala and the right amygdala by selecting all atlas voxels assigned to those regions and saving them as NIfTI images in standard MNI space. The dmPFC ROI was defined as a spherical region centered at MNI coordinates (0, 50, 30), approximating the midpoint of dmPFC peaks reported by Cremers et al. (2010) and Deng et al. (2019). We generated a 10 millimeter radius sphere around this coordinate in 2 millimeter MNI space, with the amygdala atlas image as a reference for affine and grid dimensions. A 10 millimeter radius was chosen to balance anatomical specificity and individual variability in dmPFC anatomy. The resulting binary mask was saved as a NIfTI image.

For each participant and contrast, we summarized first-level effects within each ROI. We applied a fitted NIfTImasker to the beta map of each subject and averaged voxel-wise effect estimates across all voxels within a given ROI, yielding one mean beta value per subject, per contrast, and per ROI.

Second-level Analyses

Second-level analyses related between-subject variability in neural responses to individual differences in neuroticism, while controlling for relevant demographic and motion covariates. We conducted both ROI-based analyses on extracted mean contrast estimates and whole-brain voxelwise regressions on individual contrast maps. All second-level models included neuroticism as the main predictor, and age, sex, and mean FD as covariates when available.

ROI-level Regression Models

For each ROI and each of the four first-level contrasts, we modeled mean beta values across subjects using ordinary least squares regression. The dependent variable was the mean contrast estimate within a ROI for a specific contrast and subject. Predictor variables were

drawn from the combined ROI and phenotype labels: neuroticism (z-scored), age, sex, and mean FD were included as covariates where present. Models were estimated using ordinary least squares (OLS) as implemented in the OLS class from the statsmodels Python package (Seabold & Perktold, 2010). We recorded the regression coefficient, t-value, and p-value for neuroticism, along with sample size and model R^2 , for each ROI-contrast combination.

The ROI-level estimates were then subjected to multiple-comparisons correction using the Benjamini-Hochberg false discovery rate (FDR) procedure with a nominal alpha of 0.05. Since most contrasts had directional hypotheses, we first converted the two-sided p-values for neuroticism into one-sided p-values for H01, H02, and H04, while retaining two-sided p-values for H03. P-values were submitted to the FDR procedure, yielding FDR-adjusted p-values and a binary indicator of significance for each ROI-contrast combination.

Whole-Brain Voxelwise Analysis

We performed voxelwise second-level regression for each contrast using Nilearn's SecondLevelModel class (Abraham et al., 2014). For each contrast, we gathered the corresponding first-level beta maps across all subjects with complete phenotype data, and constructed a design matrix with z-scored neuroticism as the predictor of interest and age, sex, and mean FD as covariates. Subjects with missing values in any of these predictors were excluded from the analysis of that contrast. Contrasts with less than 10 valid subjects were not analyzed. The model was fitted separately for each contrast using the MNI152 2 millimeter brain mask as a group-level mask, and then computed a z-statistic map for the neuroticism regressor. For each contrast, we saved the unthresholded z-maps in NIfTI format and generated glass-brain visualizations for descriptive purposes.

We attempted to identify clusters showing FDR-controlled effects of neuroticism by thresholding the voxelwise z-maps using FDR with alpha 0.05. When successful, the resulting FDR-corrected maps were saved and visualized. As a supplementary, more

exploratory analysis, we produced thresholded maps using an uncorrected voxelwise threshold of $p < 0.001$ combined with a cluster-extent threshold of 20 voxels. Both the FDR-corrected and the $p < 0.001$ cluster-extent maps were saved as NIfTI images, and corresponding glass-brain figures were generated to help interpretation.

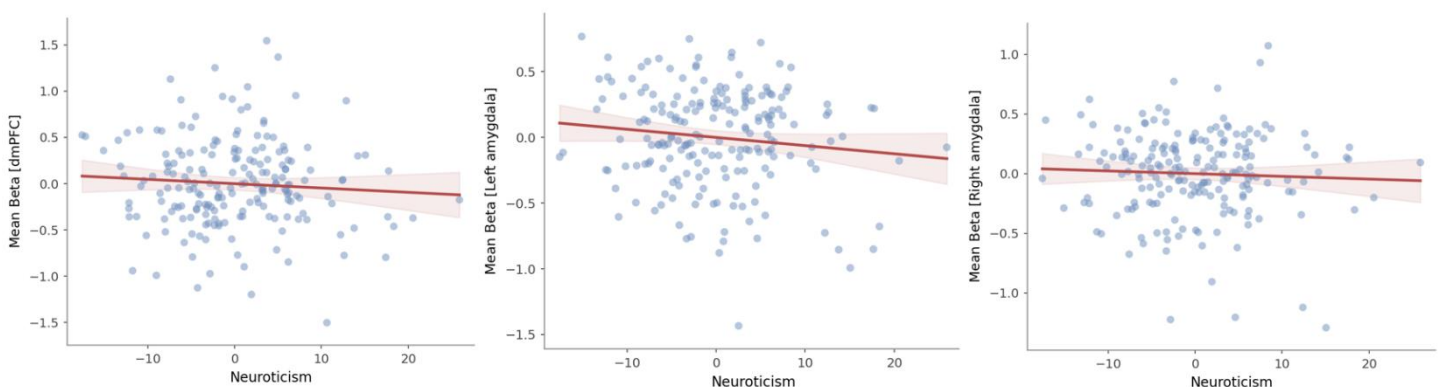
Results

Emotional Expressions versus Neutral Expressions

H01 predicted that higher trait neuroticism would be associated with greater activation in the bilateral amygdala and dmPFC for all emotional expressions (anger, contempt, joy, and pride combined) relative to neutral expressions. However, none of the three ROIs yielded a significant positive association after FDR correction (see Figure 1). All three beta coefficients were negative, indicating a non-significant tendency toward lower ROI activation with increasing neuroticism. The left amygdala showed $\beta = -0.047$, with $t(191) = -1.69$, $p_{\text{unc}} = .093$, and $p_{\text{FDR}} = .964$; the right amygdala $\beta = -0.017$, with $t(191) = -0.66$, $p_{\text{unc}} = .509$, and $p_{\text{FDR}} = .964$; the dmPFC $\beta = -0.035$, with $t(191) = -1.01$, $p_{\text{unc}} = .316$, and $p_{\text{FDR}} = .964$.

Figure 1

Scatterplots for dmPFC and bilateral amygdala for emotion vs. neutral contrasts



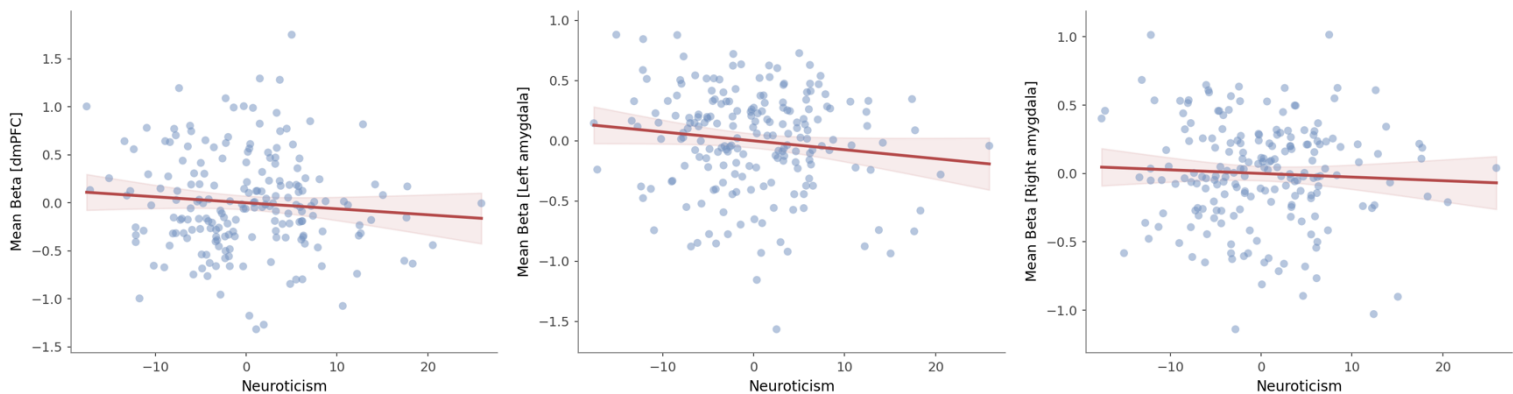
Note. The x-axis shows z-scored neuroticism; the y-axis shows residualized beta values after partialling out age, sex, and mean framewise displacement (FD).

Negative Expressions versus Neutral Expressions

H02 predicted increased amygdala and dmPFC activation for negative emotional expressions (anger and contempt) relative to neutral expressions as a function of neuroticism. No ROI reached significance after FDR correction (see Figure 2). The left amygdala yielded the numerically largest effect across all 12 tests ($\beta = -0.056$, $t(191) = -1.81$, $p_{\text{unc}} = .072$), but this effect was small, opposite to predicted direction, and did not survive FDR correction. Both the right amygdala ($\beta = -0.020$, $t(191) = -0.72$, $p_{\text{unc}} = .472$) and the dmPFC ($\beta = -0.047$, $t(191) = -1.25$, $p_{\text{unc}} = .214$ (all $p_{\text{FDR}} = .964$)) showed a non-significant effect in the hypothesized direction.

Figure 2

Scatterplots for dmPFC and bilateral amygdala for negative vs. neutral contrasts



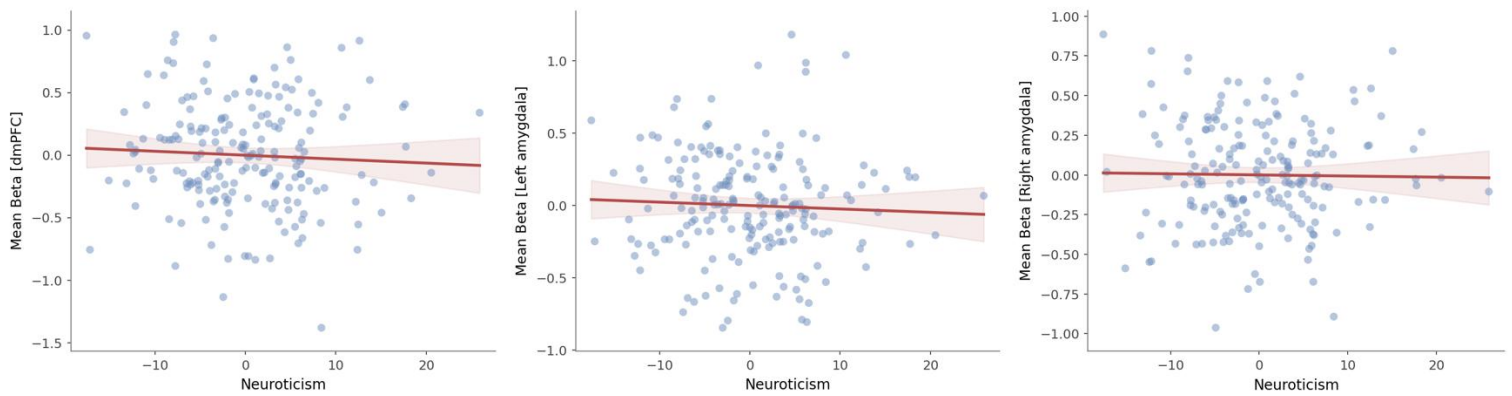
Note. The x-axis shows z-scored neuroticism scored; the y-axis shows residualized beta values after partialling out age, sex, and mean FD.

Positive Expressions versus Neutral Expressions

H03 hypothesized that neuroticism would be associated with altered (in either direction) amygdala and dmPFC responses to positive emotional expressions (joy and pride) relative to neutral. No significant associations were found in either direction after FDR correction (see Figure 3). Left amygdala showed $\beta = -0.038$, with $t(191) = -1.24$, $p_{\text{unc}} = .218$, and $p_{\text{FDR}} = .964$; right amygdala $\beta = -0.015$, $t(191) = -0.49$, $p_{\text{unc}} = .624$, and $p_{\text{FDR}} = .964$; dmPFC $\beta = -0.023$, with $t(191) = -0.60$, $p_{\text{unc}} = .551$, and $p_{\text{FDR}} = .964$.

Figure 3

Scatterplots for dmPFC and bilateral amygdala for positive vs. neutral contrasts



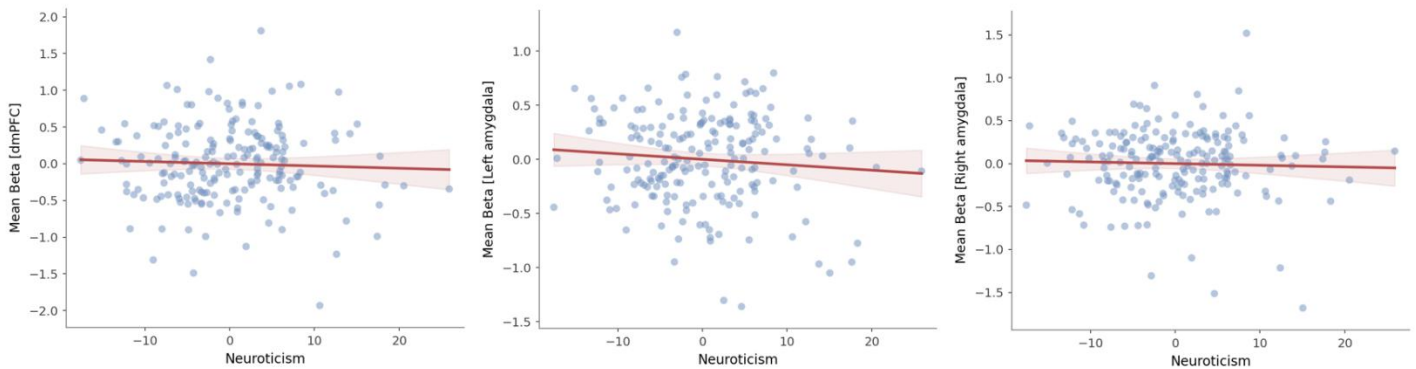
Note. The x-axis shows z-scored neuroticism; the y-axis shows residualized beta values after partialling out age, sex, and mean FD.

Negative versus Positive Expressions

H04 tested the directional hypothesis that the neuroticism-ROI association would be stronger for negative than for positive expressions. No significant effects were found in any ROI after FDR correction (see Figure 4). Left amygdala showed $\beta = -0.018$, with $t(191) = -0.66$, $p_{\text{unc}} = .508$, and $p_{\text{FDR}} = .964$; right amygdala $\beta = -0.005$, with $t(191) = -0.22$, $p_{\text{unc}} = .829$, and $p_{\text{FDR}} = .964$; dmPFC $\beta = -0.024$, with $t(191) = -0.75$, $p_{\text{unc}} = .452$, and $p_{\text{FDR}} = .964$. Table 1 presents the complete set of ROI regression results for all four contrasts and three ROIs.

Figure 4

Scatterplots for dmPFC and bilateral amygdala for negative vs. positive contrasts



Note. The x-axis shows z-scored neuroticism; the y-axis shows residualized beta values after partialling out age, sex, and mean FD.

Table 1

ROI Regression Results: Neuroticism Predicting Mean BOLD Activation (N = 196)

Contrast	ROI	β	$t(191)$	p (unc.)	p (FDR)
Emotion vs. Neutral	Left amygdala	-0.047	-1.69	.093	.964
	Right amygdala	-0.017	-0.66	.509	.964
	dmPFC	-0.035	-1.01	.316	.964
Negative vs. Neutral	Left amygdala	-0.056	-1.81	.072	.964
	Right amygdala	-0.020	-0.72	.472	.964
	dmPFC	-0.047	-1.25	.214	.964
Positive vs. Neutral	Left amygdala	-0.038	-1.24	.218	.964
	Right amygdala	-0.015	-0.49	.624	.964
	dmPFC	-0.023	-0.60	.551	.964
Negative vs. Positive	Left amygdala	-0.018	-0.66	.508	.964
	Right amygdala	-0.005	-0.22	.829	.964
	dmPFC	-0.024	-0.75	.452	.964

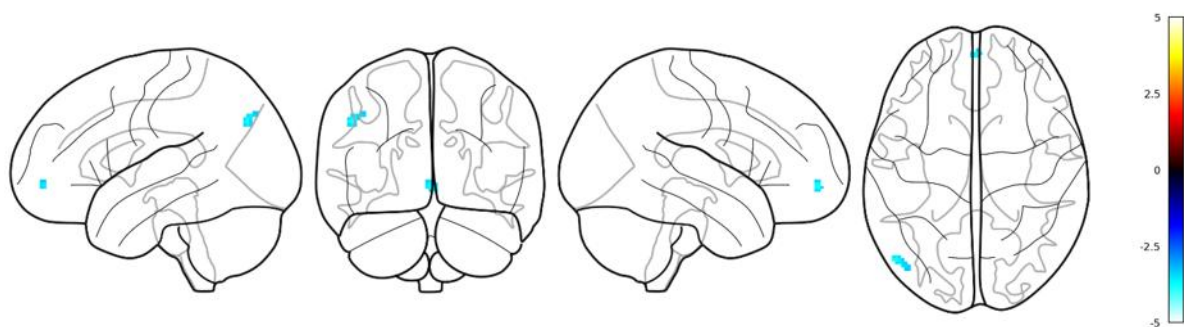
Note. β = regression coefficient for z-scored neuroticism; p (unc.) = uncorrected p -value (one-tailed for H01, H02, H04; two-tailed for H03); p (FDR) = Benjamini-Hochberg FDR-corrected p -value across all 12 tests simultaneously. dmPFC = dorsomedial prefrontal cortex.

Whole-Brain Voxelwise Analysis

Whole-brain second-level regression of neuroticism on individual first-level contrast maps was performed for all four contrasts. The resulting statistical maps were thresholded at a Benjamini-Hochberg FDR threshold of $p < .05$. No voxels survived this correction for any contrast. As an exploratory analysis, whole-brain maps were additionally thresholded at an uncorrected $p < .001$ with a minimum cluster extent of 20 contiguous voxels. Small suprathreshold clusters emerged in three of the four contrasts. For emotion-versus-neutral (Figure 5) and negative-versus-neutral (Figure 6), exploratory maps revealed small suprathreshold clusters, mainly in the negative direction (lower activation with higher neuroticism), distributed across frontal and temporal regions. For negative-versus-positive (Figure 7), clusters in both directions were present. The positive-versus-neutral contrast yielded no suprathreshold clusters at this exploratory threshold, thus no uncorrected map is shown for that contrast. These exploratory results are reported for completeness only. Given the absence of any effect surviving FDR correction, they should not be interpreted as confirmatory evidence for any of the study hypotheses.

Figure 5

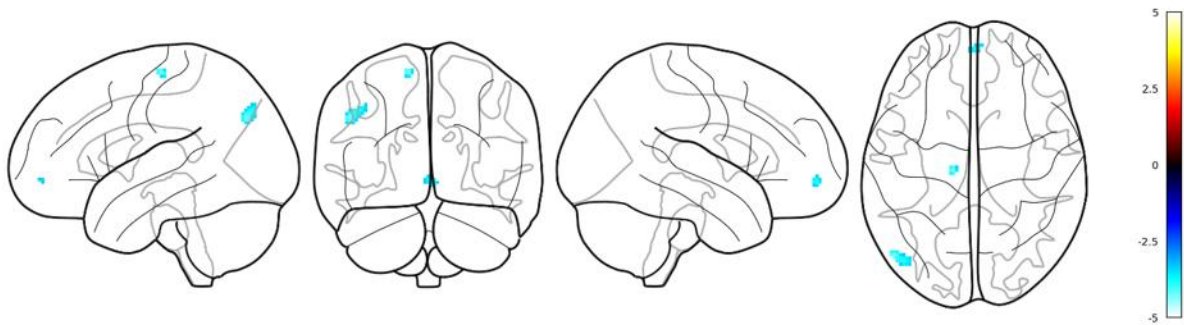
Exploratory Glass Brain Map for the Emotion-versus-Neutral Contrast



Note. Statistical maps display regions showing significant associations between neuroticism and neural activation during the emotion-versus-neutral contrast ($p < .001$, uncorrected; minimum cluster extent $k \geq 20$ voxels).

Figure 6

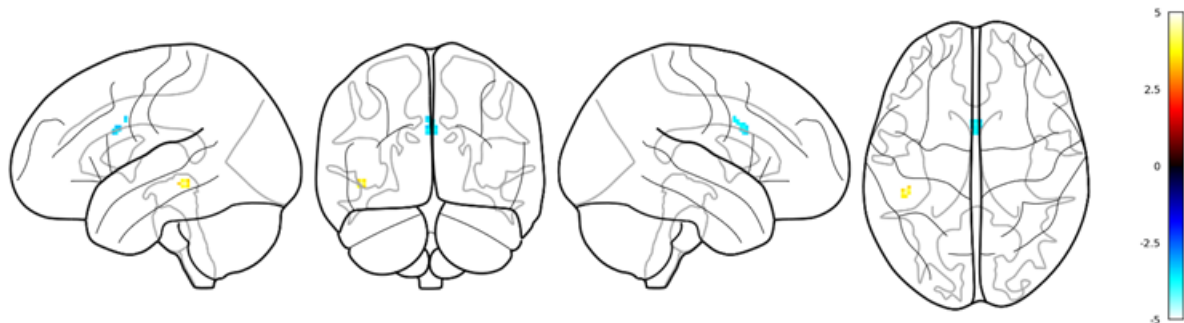
Exploratory Glass Brain Map for the Negative-versus-Neutral Contrast



Note. Statistical maps display regions showing significant associations between neuroticism and neural activation during the negative-versus-neutral contrast ($p < .001$, uncorrected; minimum cluster extent $k \geq 20$ voxels).

Figure 7

Exploratory Glass Brain Map for the Negative-versus-Positive Contrast



Note. Statistical maps depict brain regions showing significant associations between neuroticism and neural activation during the negative-versus-positive contrast ($p < .001$, uncorrected; minimum cluster extent $k \geq 20$ voxels).

Discussion

The present study examined whether trait neuroticism was associated with altered BOLD activation in the bilateral amygdala and dmPFC during a dynamic emotional face perception task. Four hypotheses were tested, each addressing a progressively more specific prediction about the direction and valence-specificity of these associations. None of the four hypotheses were supported by the ROI analyses, and no voxels survived FDR correction in the whole-brain analyses. The following sections discuss the implications of these null results for each hypothesis in turn, consider potential explanations for the absence of significant findings, and reflect on the strengths and limitations of the current study.

Emotional Expressions versus Neutral Expressions

The first hypothesis predicted that higher neuroticism scores would be associated with greater amygdala and dmPFC activation for emotional expressions relative to neutral, reflecting a generally heightened neural sensitivity to emotionally loaded faces. This prediction was not supported. If anything, the observed beta coefficients pointed in the

negative direction across all three ROIs. This pattern differs from findings suggesting that neuroticism is associated with heightened neural responses to emotional stimuli (Stein et al., 2007; Chan et al., 2009). Prior research has also reported null findings, with several studies failing to detect significant associations between neuroticism and amygdala or dmPFC activity (Klamer et al., 2017; Silverman, 2018), suggesting that any association between neuroticism and regional BOLD responses to emotional faces is likely small and context-dependent.

Negative Expressions versus Neutral

The second hypothesis predicted that neuroticism would be associated with increased amygdala and dmPFC activation in response to negative facial expressions (anger and contempt) relative to neutral expressions. This prediction was based on evidence that higher neuroticism is associated with increased sensitivity to negative emotional information. This includes faster detection and a lower threshold for perceiving faces as negative (Chan et al., 2007; Hopkins et al., 2021), as well as greater amygdala and dmPFC activation during negative face processing (Brown et al., 2020; Chan et al., 2009; Cremers et al., 2010; Stein et al., 2007). However, this hypothesis was not supported. Given that several previous studies have likewise failed to detect significant associations between neuroticism and amygdala or dmPFC activity during emotional face processing (Klamer et al., 2017; Meiering et al., 2026; Silverman, 2018), the present null finding may reflect the broader inconsistency of the literature. Although the left amygdala showed a slight negative effect ($\beta = -0.056$, $p = .072$), we refrain from interpreting this post hoc given its small magnitude and non-significance after FDR correction. If such an effect exists, it could be compatible with suppressive coping in highly neurotic individuals (Chen et al., 2017), but that interpretation remains speculative because the result was non-significant and uncorrected.

Positive Expressions versus Neutral

The third hypothesis was non-directional, predicting that neuroticism would be associated with altered amygdala and dmPFC activation to positive facial expressions (joy and pride) relative to neutral. This prediction was motivated by the mixed evidence in the literature. While some studies have reported increased amygdala responses to positive faces in anxiety-prone individuals (Stein et al., 2007), others have found no association between neuroticism and neural responses to positive stimuli (Haas et al., 2008; Klamer et al., 2017). The present study found no significant effects in either direction. Although behavioral work suggests that highly neurotic individuals may process positive emotional expressions less efficiently (Andric et al., 2016; Sawada et al., 2016), these behavioral differences do not appear to be reflected in differential amygdala or dmPFC activation.

Negative versus Positive Expressions

The fourth and final hypothesis predicted that the association between neuroticism and ROI activation would be stronger for negative than for positive facial expressions. This valence-specificity prediction was not supported, providing no evidence that neuroticism is associated with a negativity bias in amygdala or dmPFC activation. The lack of a valence-specific effect in the amygdala or dmPFC is not necessarily inconsistent with the literature. Klamer et al. (2017) found valence-specific effects in the left MTG, with stronger associations between neuroticism and activation for negative compared to positive facial expressions. This suggests that valence-specific effects may be mediated by processes occurring outside the ROIs examined in the present study.

Possible Explanations for Null Results

Several factors may account for the overall absence of significant associations across all four hypotheses. It is important to consider the nature of the stimuli used in the face perception task. The ADFES stimuli used in this paradigm are dynamic videos depicting

transitions from neutral to full emotional expressions. This is a more ecologically valid format than static photographs used in prior studies linking neuroticism to amygdala and dmPFC activation (Chan et al., 2009; Cremers et al., 2010; Brown et al., 2020). However, dynamic and static faces are processed differently in the brain. Dynamic faces engage a broader network of regions involved in motion processing and social perception (Kessler et al., 2011). This broader engagement may have weakened the neuroticism-related effects previously observed with static stimuli. As a result, effects may have been distributed across a wider network rather than being detectable in the amygdala or dmPFC.

The present null results are consistent with the possibility that the effects of neuroticism on emotional face processing are better captured at the level of functional connectivity rather than regional activation. This is consistent with Meiering et al. (2026), who reported that neuroticism was associated with altered connectivity across a broad network, including the amygdala, dorsolateral prefrontal cortex, anterior insula, and dorsal cingulate cortex, without corresponding effects on regional activation levels. Similarly, Cremers et al. (2010) found that amygdala-dmPFC connectivity was modulated by neuroticism. The present study focused on univariate activation and was not designed to test connectivity or multivariate patterns. Together with earlier work linking neuroticism to changes in connectivity rather than regional activation, our findings show the need for studies that combine ROI-based and network-level approaches to characterize neuroticism-related neural effects.

Conclusion

Overall, in a large sample of healthy young adults from the PIOP1 dataset, we found no robust associations between trait neuroticism and activation in the bilateral amygdala or dmPFC during a dynamic emotional face perception task. This suggests that any valence-specific hyperreactivity in these regions as a central neural signature of neuroticism is likely

small, task-dependent, or reflected in network interactions rather than regional activity.

Future studies that combine adequate powered samples with connectivity- and pattern-based analyses, and with a broader range of emotions, will be crucial for understanding how neuroticism influences neural processing of facial emotion.

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